INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

SALTLAKE

(School of University of Engineering and Management, Kolkata)

End Semester Examination, December - 2024

Degree: **B.C.A**, Semester: **First**

Stream: **BCA**

Paper Code: **BCACC103**

Paper Name: **Discrete Structure**

Full Marks:100 Duration:3 Hours

Part - A

Attempt 10 questions out of 15 questions

Each question carries 2 Marks (2 X 10)

1. If P = { x : x is an integer and $x \geq 6$}. 2

Q = { x : x is an integer and $-3 \leq x \leq 10$} then find

$P \cup Q$, $P \cap Q$ and verify that $(P \cup Q) \cap (P \cup Q) = P$.

2. Prove that $(A \cap C) \cup (B \cap C') = \emptyset \Longrightarrow A \cap B = \emptyset$. 2

3. Prove that $A \cap B = \emptyset$ if and only if $A - B = A$. 2

4. If f, g : R → R where $f(x) = ax + b$, $g(x) = 1 - x + x^2$ and $(g \circ f)(x) = 9x^2 - 9x + 3$, find the values of a and b. 2

5. What is the number of elements in the power set of {a, b, c, d} ? 2

6. There are four vertices P, Q, R, S. There are five pairs (x, y) where the vertex x is connected with vertex y. Find a set of edges. Show the graph. 2

7. Find the minimum and maximum number of edges of a simple graph with 10 vertices and 3 components. 2

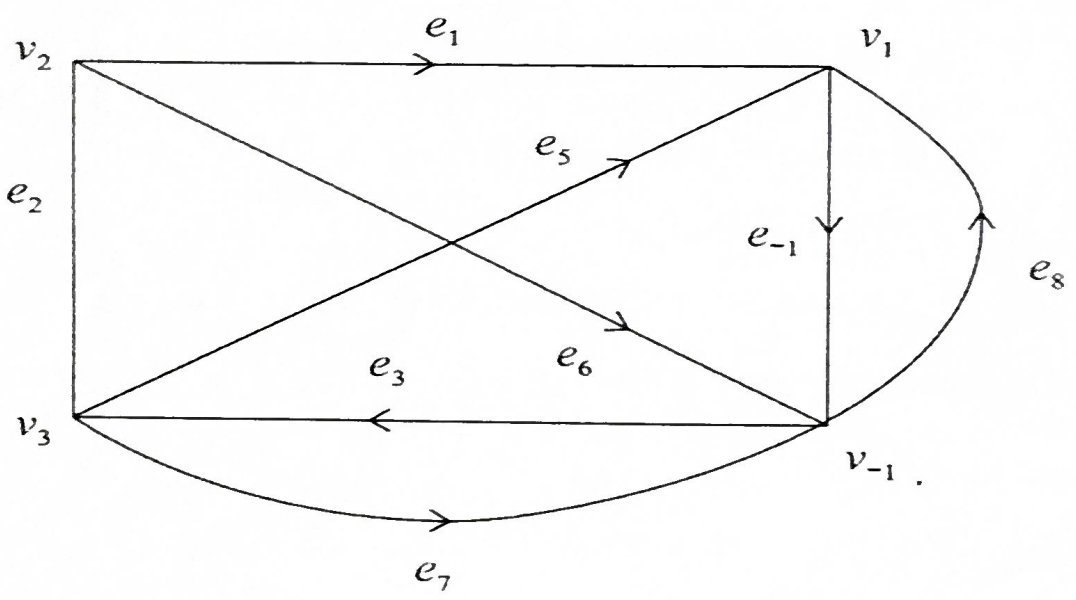
8. Give an example of a Trail which is not a walk. 2

9. Draw a connected graph having six vertices, five edges and a circuit. 2

10. Define spanning tree of a connected graph. Exhibit this for a graph. 2

11. Draw a tree with six vertices having degree 1, 1, 1, 1, 3, 3. 2

12. Write the incidence matrix of the following di-graph: 2



13. Prove that $\rightarrow (P \vee Q) \vee (\rightarrow P \wedge Q) \vee P$ is a tautology. 2

14. Let p be a proposition 'He is intelligent' and q be a proposition 'He is tall'. Write the following proposition in symbolic form: 2

i. He is intelligent and tall
ii. He is tall or he is not tall and intelligent.

15. Prove that the following propositions are tautology: 2
$[p \wedge (p \rightarrow q)] \rightarrow q$

Part - B

Attempt 6 questions out of 9 questions
Each question carries 5 Marks (5 X 6)

16. In an examination, out of 90 students 65 passed in Physics and 50 passed in Mathematics and 35 passed in both Physics and Mathematics. **5**

    a) How many students passed in exactly one of the two subjects?
    b) How many students failed in both the subjects?

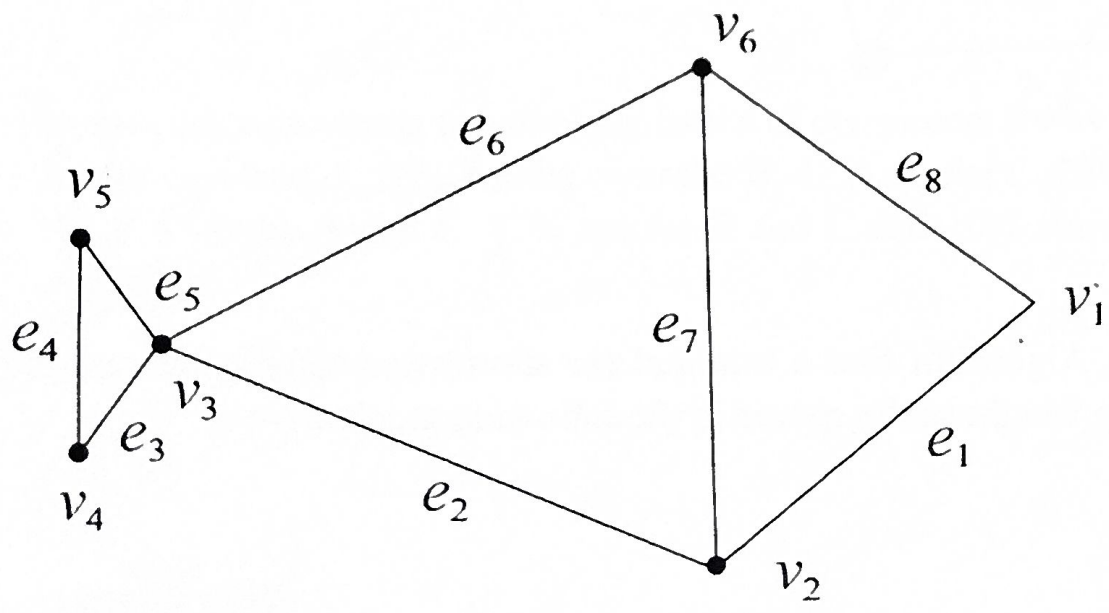
17. Let A, B, C be subsets of a Universal set U. Prove that **5**

    i. $A \times (B \cup C) = (A \times B) \cup (A \times C)$.
    ii. $A \times (B - C) = (A \times B) - (A \times C)$.

18. Determine whether the function $f : R \to R$ defined by $f = \{(x, y) : y = x^3\}$ is invertible and if so, determine f^{-1}. **5**

19. Draw, if possible, a simple graph with five vertices having degrees 2, 3, 3, 3, 3. **5**

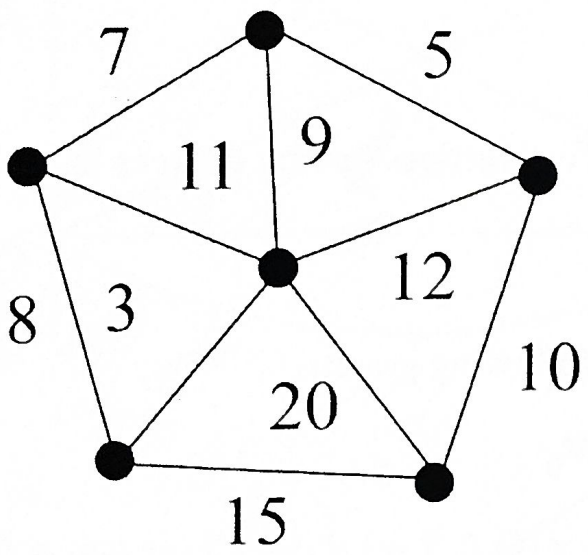
20. Find three circuits from V_3 to V_3 in the following graph: **5**



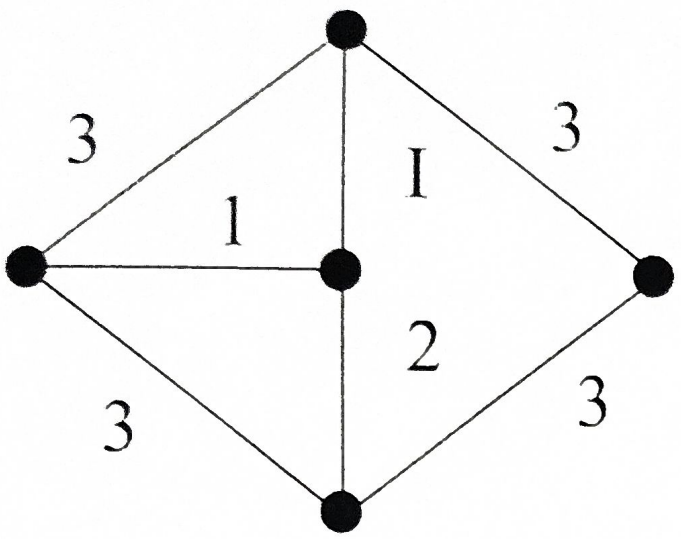
21. Draw the graph whose incidence matrix is **5**

$$\begin{bmatrix} 0 & 0 & 1 & -1 & 1 \\ -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 \end{bmatrix}$$

22. Find by Kruskal's Algorithm, a minimal spanning tree of the following weighted graph: Also calculate total weight of the obtained spanning tree 5



23. By Kruskal's Algorithm find a minimal (or shortest) spanning tree and the corresponding weight of the sp tree in the following graphs: 5



24. Show that $\{(p \wedge \sim q) \rightarrow r\} \rightarrow \{p \rightarrow (q \vee r)\}$ is a tautology. **5**

Part - C

Attempt 5 questions out of 8 questions
Each question carries 10 Marks (10 X 5)

25. If A and B are sets, then prove that A – B , A ∩ B and B – A are pairwise disjoint. **10**

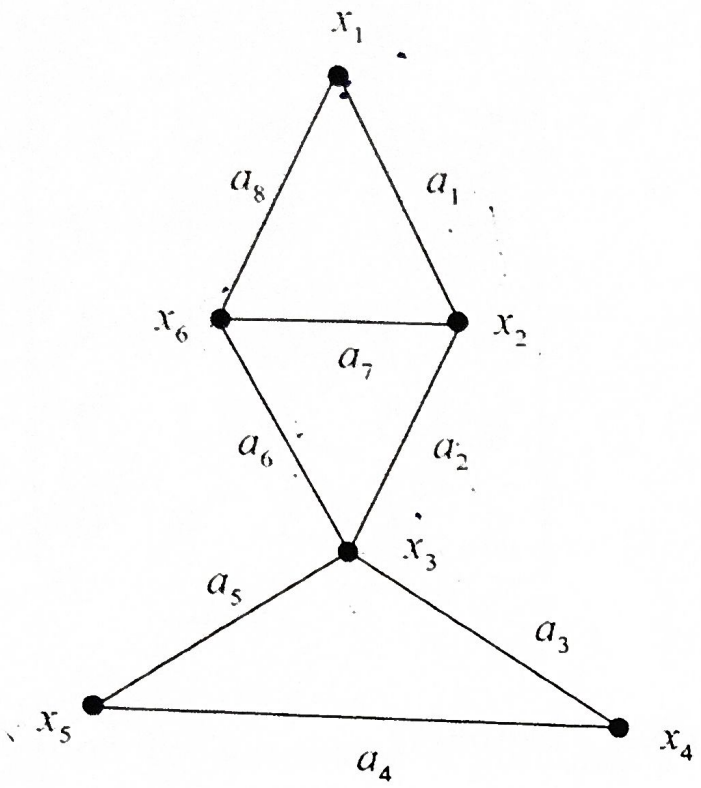
26. Prove that **10**

i. $A \cup B = A \cap B \Longrightarrow A = B$.
ii. $(A \Delta B') = (A' \cup B) \cap (A \cup B')$.

27. In a survey concerning the smoking habits of consumers it was found that 55% smoke cigarette A, 50% smoke cigarette B, 42% smoke C, 28% smoke A and B, 20% smoke A and C, 12% smoke B and C and 10% smoke all the three cigarettes. **10**

i. What percentage do not smoke?
ii. What percentage smoke exactly 2 brands of cigarettes?

28. From the graph: **10**



You are given some walks. Determine which are a path, a trail, a closed walk, a circuit;

i. $x_1, a_1, x_2, a_2, x_3, a_5, x_5, a_4, x_4$
ii. $x_2, a_7, x_6, a_6, x_3, a_3, x_4, a_4, x_5, a_5, x_3$
iii. $x_6, a_7, x_2, a_2, x_3, a_3, x_4, a_4, x_5, a_5, x_3, a_6, x_6$
iv. $x_1, a_1, x_2, a_2, x_3, a_6, x_6$
v. $x_2, a_2, x_3, a_6, x_6, a_8, x_1, a_1, x_2$

29. Construct the graph or digraph corresponding to the following incidence matrix **10**

(i) $\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$ (ii) $\begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & -1 & 0 & -1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ (iii) $\begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 \end{bmatrix}$

30. Find the graph or Di-graphs whose Adjacency matrices are given below. Find which one of these are disconnected: **10**

i) $\begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$

ii) $\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$

31. The following table shows the distances, in kilometers, between six villages in India. Find a minimal spanning tree connecting the six villages applying Kruskal's Algorithm: **10**

..	*A*	*B*	*C*	*D*	*E*	*F*
A	-	5	6	12	4	7
B	5	-	11	3	2	5
C	6	11	-	8	6	6
D	12	3	8	-	7	9
E	4	2	6	7	-	8
F	7	5	6	9	8	-

32. Show that **10**

$p \wedge (q \leftrightarrow r) \vee (r \leftrightarrow p) \equiv p \wedge \{(q \rightarrow r) \wedge (r \rightarrow q)\} \vee \{(r \rightarrow p) \wedge (p \rightarrow r)\}$

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